MCMC notes

↳ Idea: Sample from posterior state $\propto$ likelihood · prior

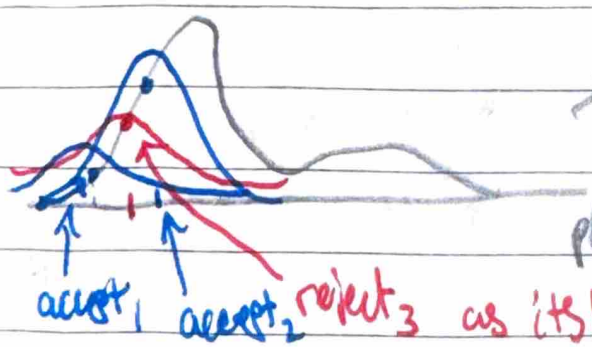
↓

in Stan lts

$\log(\text{posterior}) \propto \log(L(x_1, \dots, x_n \mid \text{prior})) + \log(\text{prior})$

might switch to variational inference later when need to scale

$$\frac{f(x')}{f(x)} = \frac{\tilde{f}(x')/\cancel{Z}}{\tilde{f}(x)/\cancel{Z}} = \frac{\tilde{f}(x')}{\tilde{f}(x)}$$ } when high we explore higher probability



→ random walk when it proposes a new θ from plausible sample space ⇒ compares to log(likelihood) + log(prior)

↓

moves or stays decided on that

↓

do this lots of times

↓

histogram ≈ posterior state

$$K(x, x') = \begin{cases} q(x' \mid x) \min\left(1, \frac{\tilde{f}(x')}{\tilde{f}(x)}\right), & x \neq x' \\ 1 - \int q(x' \mid x) \min\left(1, \frac{\tilde{f}(x')}{\tilde{f}(x)}\right) dx', & x = x' \end{cases}$$

$K(\theta, \theta')$ = prob. of moving from $\theta \to \theta'$

$\pi(\theta) = p(\theta \mid \text{data})$

proposal dist. := $q(\theta' \mid \theta)$

accept prob. := $a(\theta, \theta') = \min\left(1, \frac{\pi(\theta')\, q(\theta \mid \theta')}{\pi(\theta)\, q(\theta' \mid \theta)}\right)$

proportion of

$\frac{\text{likelihood x prior of } \theta'}{\text{likelihood x prior of } \theta}$

moving vs staying :=

prob of moving to proposed parameter value

$$K(\theta, \theta') = \begin{cases} q(\theta' \mid \theta)\, a(\theta, \theta'), & \theta' \neq \theta \\ 1 - \int q(\theta' \mid \theta)\, a(\theta, \theta')\, d\theta', & \theta' = \theta \end{cases}$$

total prob of moving somewhere else

Method of prior

$p(\text{data}|\theta)\,p(\theta)$

$p(\theta' \to \theta) = p(\theta \to \theta')$ reversibility $\Rightarrow$ stationary

this forces

$\Rightarrow$ $\pi(\theta)$ needs to be stationary! $\Rightarrow$ violates V2

Thus we need $\pi(\theta)\,K(\theta, \theta') = \pi(\theta')\,K(\theta', \theta)$

since $K(\theta, \theta') = \underbrace{q(\theta|\theta')}_{\text{proposal dist}}\,\underbrace{(a(\theta, \theta'))}_{\text{acceptance}}$, $\theta \neq \theta'$

$$\pi(\theta)\,q(\theta|\theta')\,a(\theta, \theta') = \pi(\theta')\,q(\theta'|\theta)\,a(\theta', \theta)$$

$$a(\theta, \theta') = \min\left(1, \frac{\pi(\theta')\,q(\theta'|\theta)}{\pi(\theta)\,q(\theta|\theta')}\right)$$

$$\pi(\theta)\,q(\theta'|\theta)\,a(\theta, \theta') = \pi(\theta')\,q(\theta|\theta')\,a(\theta', \theta)$$

$$\Rightarrow a(\theta, \theta') = \min\left(\underbrace{\frac{\pi(\theta')\,q(\theta|\theta')}{\pi(\theta)\,q(\theta'|\theta)}}_{L \text{ fun}}, 1\right)$$

if $L \geq 1$

$\Rightarrow$ accept

if $L < 1$ $\Rightarrow$ accept sometimes depends on probability

MY MCMC

State-space := $\{\lambda\}\{\sigma, \varepsilon\}$ multi-variate normal dist

random-walk := $q(\theta', \theta) = N(\theta, \Sigma)$

parameter current value — step-size

acceptance := $a(\theta, \theta') = \min\left(1, \frac{\pi(\theta')}{\pi(\theta)}\right)$

no $q(\theta|\theta')$, $q(\theta'|\theta)$ because im assuming symmetric.

$$\Sigma = \begin{array}{c} \begin{array}{ccc} \lambda & \sigma & \varepsilon \end{array} \\ \begin{pmatrix} V_1 & CV & CV \\ CV & V_2 & CV \\ CV & CV & V_3 \end{pmatrix} \end{array}$$

CV = covariance
V = variance of parameter

$$\Sigma = \begin{pmatrix} Var(\lambda) & 0 & 0 \\ 0 & Var(\sigma) & cov(\sigma, \varepsilon) \\ 0 & cov(\sigma, \varepsilon) & var(\varepsilon) \end{pmatrix}$$

as $\{\lambda\}$ independent of $\{\sigma, \varepsilon\}$

$$\Sigma_{\{\lambda\}} = \left(Var(\lambda)\right)$$

$$\Sigma_{\{\sigma, \varepsilon\}} \begin{pmatrix} Var(\sigma) & cov(\sigma, \varepsilon) \\ cov(\sigma, \varepsilon) & Var(\varepsilon) \end{pmatrix}$$

as VI assumes stationarity
$\pi(\theta) K(\theta, \theta') = \pi(\theta') K(\theta', \theta)$
↓
in an HMC
it means that it is 'time-reversal'.

in my Stan it uses HMC

- chains = 4 => 4 different starting points, paths are independent of one another; goal => all 4 converge to same values
- iter_sampling = 2000 => 2000 × 4 = 8000 total samples
- iter_warmup = 1000 => builds the mass matrix (Σ) ← 2 of them and gets rid of unreliable values in state-space $\{\lambda, \sigma, \varepsilon\}$

my Hmc creates 2 (Σ) matrix because β's and GPD are independent
↓
Propose a linked parameter maybe 'Hmc chains'? => would that help Σ?
↓
allow for Hmc to sample for more correlation in $\{\sigma, \varepsilon\}$ matrix "dense_e" => more computational cost but more accurate

$$\Sigma_{\{\sigma, \varepsilon\}} \begin{pmatrix} \frac{1}{Var(\sigma)} & cov(\sigma, \varepsilon) \\ cov(\sigma, \varepsilon) & \frac{1}{Var(\varepsilon)} \end{pmatrix}$$

=> $cov(\sigma, \varepsilon)$ is 0 rn but if I use dense_e it will adjust and find cov structure